

Distributed upcycling of mixed plastic wastes: material challenges and opportunities for additive manufacturing.

Plastic waste is silent crisis that increasingly threatens planetary boundaries (Villarrubia-Gómez et al. 2024) and the safe operating space for humanity. Distributed recycling for additive manufacturing (DRAM) offers a pathway to reduce manufacturing costs by transforming plastic waste into 3D printing feedstock.

This study evaluates the geographical reproducibility of a distributed recycling and open-source manufacturing approach originally developed in France. The approach enables the local transformation of post-consumer plastic bottles into new products through large-scale direct screw-feeding additive manufacturing, when implemented in a North American context. The results show that the operational difficulty index (ODI) increased from 1.6 to 2.2, highlighting challenges related to material collection, shredding, and printing. Different recycled feedstocks were assessed in terms of particle characteristics, extrusion behavior, and printed-part quality. Process parameter optimization improved material flow consistency and printability despite the irregular geometry of recycled particles. Mechanical testing demonstrated that recycled blend materials compatibilized with 5 wt % SEBS-G1652 achieved tensile strengths of 19 ± 3.5 MPa, demonstrating the mechanical performance attainable with the compatibilized recycled material.. Finally, the potential use of recycled feedstock was demonstrated by 3D printing large-scale parts as a case study of real-world applications.

Current centralized waste management systems remain structurally inadequate (Lerpiniere, Wilson, and Velis 2025) to address the complex and systemic nature of plastic pollution.

Recently, advances in upcycling approaches, both mechanical(Hassan, Mohanty, and Misra 2024) and chemical(Lepage and Gras 2026), are emerging as a promising strategy to valorize plastic waste by transforming it into products of higher value.

Hassan, Malik, Amar K. Mohanty, and Manjusri Misra. 2024. “3D Printing in Upcycling Plastic and Biomass Waste to Sustainable Polymer Blends and Composites: A Review.” *Materials & Design* 237 (January): 112558. <https://doi.org/10.1016/j.matdes.2023.112558>.

Lepage, Mathieu L., and Emmanuel Gras. 2026. “Upcycling Incompatible Plastics.” *Nature Sustainability* 9 (2): 185–87. <https://doi.org/10.1038/s41893-025-01724-4>.

Lerpiniere, David J., David C. Wilson, and Costas A. Velis. 2025. “Official Development Finance in Solid Waste Management Reveals Insufficient Resources for Tackling Plastic Pollution: A Global Analysis of Two Decades of Data.” *Resources, Conservation and Recycling* 212 (January): 107918. <https://doi.org/10.1016/j.resconrec.2024.107918>.

Villarrubia-Gómez, Patricia, Bethanie Carney Almroth, Marcus Eriksen, Morten Ryberg, and Sarah E. Cornell. 2024. “Plastics Pollution Exacerbates the Impacts of All Planetary Boundaries.” *One Earth* 7 (12): 2119–38. <https://doi.org/10.1016/j.oneear.2024.10.017>.